

Part 2: Job Postings Analysis

Generative AI - Assignment 1 Darrsheni Sapovadia (27PGAI0063)

This dataset is a scrape of job boards, with a job title and a long description for each posting. The brief asks for the first 25. For every posting I work out a broad category for the role, then pull the required skills, education level and experience out of the description.

Same setup as Part 1: LangChain on top of Groq, using `openai/gpt-oss-120b`.

Setup

```
In [1]: import json
import os
import re
import time

import pandas as pd
from dotenv import load_dotenv
from langchain_core.output_parsers import StrOutputParser
from langchain_core.prompts import ChatPromptTemplate
from langchain_groq import ChatGroq

load_dotenv("../.env")

MODEL = os.getenv("LLM_MODEL", "openai/gpt-oss-120b")

# Same settings as Part 1. reasoning_effort is turned down because this model
# counts its own thinking against max_tokens, and on a long job description it
# will happily use the entire budget reasoning and return nothing at all.
llm = ChatGroq(model=MODEL, temperature=0, max_tokens=800, reasoning_effort="low")

print("Model:", MODEL)
```

Model: openai/gpt-oss-120b

```
In [2]: def ask(chain, values):
        """Run a chain, waiting it out if Groq rate limits us.

        The free tier allows 8,000 tokens a minute and some of these descriptions run
        to several thousand characters, so being limited is normal here. Groq puts
        the wait it wants in the error message, so I use that rather than guessing.
        """
        for attempt in range(6):
            try:
                return chain.invoke(values)
            except Exception as error:
                message = str(error)
                limited = "rate" in message.lower() or "429" in message
                if not limited or attempt == 5:
                    raise
                suggested = re.search(r"try again in ([\d.]+)s", message)
                time.sleep(float(suggested.group(1)) + 2 if suggested else 20)

        def find_json(text):
            """Grab the JSON object out of a reply and ignore anything around it."""
            found = re.search(r"\{.*\}", text, re.S)
            if not found:
                return {}
            try:
                return json.loads(found.group())
            except json.JSONDecodeError:
                return {}
```

Step 1: Load the dataset

The CSV has an unnamed index column left over from however it was saved, so I drop that and rename the other two to something tidier.

```
In [3]: jobs = pd.read_csv("../data/job_title_des.csv")

        print("Whole dataset:", jobs.shape)
        print("Columns:", list(jobs.columns))
        jobs.head(3)
```

```
Whole dataset: (2277, 3)
Columns: ['Unnamed: 0', 'Job Title', 'Job Description']
```

```
Out[3]:
```

	Unnamed: 0	Job Title	Job Description
0	0	Flutter Developer	We are looking for hire experts flutter develo...
1	1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ ...
2	2	Machine Learning	Data Scientist (Contractor)\n\nBangalore, IN\n...

```
In [4]: df = jobs.head(25).copy().reset_index(drop=True)
df = df.drop(columns=["Unnamed: 0"])
df = df.rename(columns={"Job Title": "Job_Title", "Job Description":
"Job_Description"})

print("Working set:", df.shape)
print()
print("Description length: shortest", df["Job_Description"].str.len().min(),
      "chars, longest", df["Job_Description"].str.len().max())
df.head()
```

Working set: (25, 2)

Description length: shortest 405 chars, longest 7645

```
Out[4]:
```

	Job_Title	Job_Description
0	Flutter Developer	We are looking for hire experts flutter develo...
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code(PDJ ...
2	Machine Learning	Data Scientist (Contractor)\n\nBangalore, IN\n...
3	iOS Developer	JOB DESCRIPTION:\n\nStrong framework outside o...
4	Full Stack Developer	job responsibility full stack engineer – react...

Step 2: Job category classification

I kept the category list from the brief. Worth saying up front that this particular dataset is scraped almost entirely from IT job boards, so nearly everything here is going to come back as Technology/IT. The list still earns its place because it gives the model somewhere to put a posting that is not a developer role, and `Others` is there as the fallback the brief asks for.

```
In [5]: DOMAINS = ["Technology/IT", "Finance", "Marketing", "Healthcare", "Education",
"Others"]

category_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You sort job postings into one broad domain. "
     "Reply with the domain name only, nothing else."),
    ("human",
     "Given the following job title and description, categorize the job into one of
the "
     "following domains: Technology/IT, Finance, Marketing, Healthcare, Education,
Others.\n\n"
     "Job: Staff Accountant\n"
     "Description: Prepare monthly reconciliations, assist with year end audit and "
     "maintain the general ledger.\n"
     "Domain category:"),
    ("ai", "Finance"),
    ("human",
     "Job: Backend Engineer\n"
     "Description: Build REST APIs in Python, work with PostgreSQL and deploy on
AWS.\n"
     "Domain category:"),
    ("ai", "Technology/IT"),
    ("human",
     "Job: {title}\n"
     "Description: {description}\n"
     "Domain category:"),
])

category_chain = category_prompt | llm | StrOutputParser()
```

```
In [6]: def classify_job(title, description):
    reply = ask(category_chain, {"title": title, "description": description[:1500]})
    for domain in DOMAINS:
        if domain.lower() in reply.lower():
            return domain
    # "IT" on its own is a common reply and would miss the check above
    if "tech" in reply.lower() or "it" in reply.lower():
        return "Technology/IT"
    return "Others"
```

```
In [7]: sample = df.loc[0]

print("Title:", sample["Job_Title"])
print("Predicted category:", classify_job(sample["Job_Title"],
sample["Job_Description"]))
```

Title: Flutter Developer

Predicted category: Technology/IT

Step 3: Requirements extraction

The brief allows either three separate prompts or one combined prompt that returns all three fields. I went with the combined one. It is a third of the API calls, and it means the model reads the description once and fills in all three fields from the same pass rather than three disconnected reads.

Plenty of these postings simply do not mention a degree or a number of years, so I tell the model to write `Not specified` rather than let it invent something.

```
In [8]: requirements_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You read job descriptions and pull out the requirements. "
     "Reply with JSON and nothing else. "
     "Never guess: if the description does not mention something, say \"Not specified\"."),
    ("human",
     "Extract the required skills, education level and years of experience from the job "
     "description below.\n\n"
     "Reply in this exact shape:\n"
     '{{$skills": ["..."], "education": "...", "experience": "..."}}\n\n'
     "skills: the tools, programming languages and specific abilities asked for.\n"
     "education: the minimum degree required or preferred, or \"Not specified\".\n"
     "experience: the years or level of experience asked for, or \"Not specified\".\n\n"
     "Job title: {title}\n"
     "Job description:\n{description}"),
    ])

requirements_chain = requirements_prompt | llm | StrOutputParser()
```

```
In [9]: def extract_requirements(title, description):
    reply = ask(requirements_chain, {"title": title, "description":
description[:2500]})
    data = find_json(reply)

    skills = data.get("skills", [])
    if isinstance(skills, str):
        skills = [skills]
    skills = [str(s).strip() for s in skills if str(s).strip()]

    education = str(data.get("education", "") or "").strip() or "Not specified"
    experience = str(data.get("experience", "") or "").strip() or "Not specified"

    return skills, education, experience
```

```
In [10]: skills, education, experience = extract_requirements(
        sample["Job_Title"], sample["Job_Description"]
    )

    print("Title:", sample["Job_Title"])
    print()
    print("Skills:    ", skills)
    print("Education: ", education)
    print("Experience:", experience)
```

Title: Flutter Developer

Skills: ['Flutter']
 Education: Not specified
 Experience: 1 year (Preferred)

Worth a look at the raw description for that first posting, because it is a good example of how thin some of these adverts are:

```
In [11]: print(sample["Job_Description"])
```

We are looking for hire experts flutter developer. So you are eligible this post then apply your resume.
 Job Types: Full-time, Part-time
 Salary: ₹20,000.00 - ₹40,000.00 per month
 Benefits:
 Flexible schedule
 Food allowance
 Schedule:
 Day shift
 Supplemental Pay:
 Joining bonus
 Overtime pay
 Experience:
 total work: 1 year (Preferred)
 Housing rent subsidy:
 Yes
 Industry:
 Software Development
 Work Remotely:
 Temporarily due to COVID-19

Step 4: Run the chains over all 25 postings

Two calls per posting, so fifty in total.

```
In [12]: categories = []
all_skills = []
all_education = []
all_experience = []

start = time.time()

for position, row in df.iterrows():
    title = row["Job_Title"]
    description = row["Job_Description"]

    categories.append(classify_job(title, description))

    skills, education, experience = extract_requirements(title, description)
    all_skills.append(skills)
    all_education.append(education)
    all_experience.append(experience)

    print(f"posting {position + 1} of {len(df)} done")
    time.sleep(2)

print(f"\nAll finished in {time.time() - start:.0f} seconds")
```

posting 1 of 25 done

posting 2 of 25 done

posting 3 of 25 done

posting 4 of 25 done

posting 5 of 25 done

posting 6 of 25 done

posting 7 of 25 done

posting 8 of 25 done

posting 9 of 25 done

posting 10 of 25 done

posting 11 of 25 done

posting 12 of 25 done

posting 13 of 25 done

posting 14 of 25 done

posting 15 of 25 done

posting 16 of 25 done

posting 17 of 25 done

posting 18 of 25 done

posting 19 of 25 done

posting 20 of 25 done

posting 21 of 25 done

posting 22 of 25 done

posting 23 of 25 done

posting 24 of 25 done

posting 25 of 25 done

All finished in 214 seconds

Step 5: Add the new columns

`Required_Skills` is kept as a comma separated string so the column is consistent and reads properly when it gets written out to CSV.

```
In [13]: df["Predicted_Category"] = categories
df["Required_Skills"] = ["", ".join(s) if s else "Not specified" for s in all_skills]
df["Education_Required"] = all_education
df["Experience_Required"] = all_experience

# the four new columns on their own first
df[["Job_Title", "Predicted_Category", "Required_Skills",
    "Education_Required", "Experience_Required"]].head(10)
```

	Job_Title	Predicted_Category	Required_Skills	Education_Required	Experience_Required
0	Flutter Developer	Technology/IT	Flutter	Not specified	1 year (Preferred)
1	Django Developer	Technology/IT	Python, Django, Flask, REST API development, R...	Not specified	Not specified
2	Machine Learning	Technology/IT	Python, Java, Machine Learning, Deep Learning,...	Graduate or M.Sc. in Computer Science, Mathema...	At least 3 years
3	iOS Developer	Technology/IT	Objective-C, Cocoa Touch, Core Data, Core Anim...	Not specified	Not specified
4	Full Stack Developer	Technology/IT	React, React Native, JavaScript, HTML, CSS, RE...	Computer Science or equivalent	5+ years web development, 2+ years React
5	Java Developer	Technology/IT	C#, .NET, .NET Core, HTML5, CSS3, MsSQL, MySQL...	Bachelor's Degree in Computer Science	2 years
6	Full Stack Developer	Technology/IT	Node.js, Java, NoSQL, MongoDB, Elasticsearch, ...	B.Sc degree in Computer Science, Engineering, ...	Minimum 2 years
7	JavaScript Developer	Technology/IT	ReactJS, NodeJS, Azure Functions, GraphQL, HTM...	Any graduation	3-8 years
8	DevOps Engineer	Technology/IT	Bash, Ruby, Python, Java, Puppet, Chef, Cloudi...	Not specified	Not specified
9	Software Engineer	Technology/IT	REST API, C/C++, Linux/Unix, Python, Go, Git, ...	Not specified	Minimum 7 years

The full dataframe, original columns and new ones together

```
In [14]: pd.set_option("display.max_colwidth", 45)
df
```

Out[14]:

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Edu
0	Flutter Developer	We are looking for hire experts flutter d...	Technology/IT	Flutter	
1	Django Developer	PYTHON/DJANGO (Developer/Lead) - Job Code...	Technology/IT	Python, Django, Flask, REST API developme...	
2	Machine Learning	Data Scientist (Contractor)\n\nBangalore,...	Technology/IT	Python, Java, Machine Learning, Deep Lear...	Gra Compu
3	iOS Developer	JOB DESCRIPTION:\n\nStrong framework outs...	Technology/IT	Objective-C, Cocoa Touch, Core Data, Core...	
4	Full Stack Developer	job responsibility full stack engineer – ...	Technology/IT	React, React Native, JavaScript, HTML, CS...	Con
5	Java Developer	Software Developer - Integration*\nImmedi...	Technology/IT	C#, .NET, .NET Core, HTML5, CSS3, MsSQL, ...	Bac C
6	Full Stack Developer	senior full stack developer \n- 1800026h c...	Technology/IT	Node.js, Java, NoSQL, MongoDB, Elasticsea...	B.Sc deg Sc
7	JavaScript Developer	Job Description:\n\nReactJS + NodeJs, Azu...	Technology/IT	ReactJS, NodeJS, Azure Functions, GraphQL...	
8	DevOps Engineer	Main Responsibilities and Deliverables:\n...	Technology/IT	Bash, Ruby, Python, Java, Puppet, Chef, C...	
9	Software Engineer	Overview\n\n\nBased in Silicon Valley, Ti...	Technology/IT	REST API, C/C++, Linux/Unix, Python, Go, ...	
10	Database Administrator	richardsontexasunited state bausp4 boeing...	Technology/IT	Database performance baseline development...	Technical k
11	Machine Learning	Remote, Any where\n2 - 4 years\n₹8L - ₹12...	Technology/IT	TensorFlow, OpenCV, Python, Artificial Ne...	
12	Machine Learning	Excited by Big Data, Machine Learning and...	Technology/IT	Java, Scala, C++, Perl, Python, R, Matlab...	Masters or I

	Job_Title	Job_Description	Predicted_Category	Required_Skills	Edu
13	Software Engineer	As a member of our Software Engineering G...	Technology/IT	Advanced knowledge of application, data a...	
14	Software Engineer	JOB DESCRIPTION\n\nJob Title: Software En...	Technology/IT	Object-oriented design, Java, .NET, Relat...	Bac Compi
15	Java Developer	Overview:\n\nWestern National is seeking ...	Technology/IT	JavaScript (ES6+), Node, HTML5, CSS3, SCS...	B.S. Deg Sc
16	Wordpress Developer	Experience: 2-5 years\n\nJob Location:- A...	Technology/IT	WordPress, Plugin development, PHP, HTML/...	B.Sc (CS/C
17	iOS Developer	Job Code : ID190704\n\nJob Description:\n...	Technology/IT	Native iOS development, Hybrid app develo...	BE (Compu T
18	Database Administrator	job description summary highly skilled te...	Technology/IT	Oracle Database (10g/11g/12c), Oracle Dat...	
19	DevOps Engineer	Job Summary\n\nOur mission at Style Theor...	Technology/IT	AWS, GCP, Linux, Ansible, Chef, Puppet, T...	
20	Database Administrator	country place mortgage founded 1995 subsi...	Technology/IT	Oracle database, Linux, SQL, database per...	E
21	Wordpress Developer	Position: WordPress Developer\nLocation: ...	Technology/IT	WordPress, CSS3, JavaScript, HTML5, jQuer...	Bac Comp
22	JavaScript Developer	MonetizeMore builds industry leading ad t...	Technology/IT	JavaScript, Python/Flask, PostgreSQL, Lin...	
23	Database Administrator	qualification essential qualification rel...	Technology/IT	database design, database administration,...	
24	PHP Developer	Job Summary\n**\nWe are looking for Fresh...	Technology/IT	PHP, WordPress, HTML, CSS, jQuery, Core P...	B.E./B.Tech./M

One row as JSON

Same shape as the example in the assignment brief.

```
In [15]: row = df.loc[3]

print(json.dumps({
    "Job_Title": row["Job_Title"],
    "Job_Description": row["Job_Description"][:200] + "... [excerpt]",
    "Predicted_Category": row["Predicted_Category"],
    "Required_Skills": [s.strip() for s in row["Required_Skills"].split(",")],
    "Education_Required": row["Education_Required"],
    "Experience_Required": row["Experience_Required"],
}, indent=2))

{
  "Job_Title": "iOS Developer",
  "Job_Description": "JOB DESCRIPTION:\n\nStrong framework outside of iOS is always a p
lus\n\niOS experience and generalist engineers with backgrounds in related technologies
is a plus\n\nA disciplined approach to development, d... [excerpt]",
  "Predicted_Category": "Technology/IT",
  "Required_Skills": [
    "Objective-C",
    "Cocoa Touch",
    "Core Data",
    "Core Animation",
    "Core Graphics",
    "Core Text",
    "third-party libraries",
    "APIs",
    "networking",
    "mobile network issues",
    "concurrency",
    "threading",
    "internationalization",
    "iOS frameworks",
    "mobile development lifecycle"
  ],
  "Education_Required": "Not specified",
  "Experience_Required": "Not specified"
}
```

Spot check

The brief asks to verify a few of the outputs by hand. Rather than read all 25 I looked at how often each field came back as `Not specified`, which points straight at the postings worth reading properly.

```
In [16]: print("Categories found:")
print(df["Predicted_Category"].value_counts().to_string())
print()

for column in ["Education_Required", "Experience_Required"]:
    missing = (df[column] == "Not specified").sum()
    print(f"{column}: {missing} of {len(df)} postings do not state this")
```

Categories found:
Predicted_Category
Technology/IT 25

Education_Required: 10 of 25 postings do not state this
Experience_Required: 7 of 25 postings do not state this

```
In [17]: # reading three of them back against the description to check the extraction is
sensible
for position in [1, 5, 9]:
    row = df.loc[position]
    print("-" * 70)
    print("Title:      ", row["Job_Title"])
    print("Category:   ", row["Predicted_Category"])
    print("Skills:      ", row["Required_Skills"][:160])
    print("Education:  ", row["Education_Required"])
    print("Experience:", row["Experience_Required"])
```

```
-----
Title:      Django Developer
Category:   Technology/IT
Skills:     Python, Django, Flask, REST API development, RPC, Linux, SQL, JSON, Automat
ed unit testing (PyUnit), Verbal and written communication
Education:  Not specified
Experience: Not specified
-----
```

```
Title:      Java Developer
Category:   Technology/IT
Skills:     C#, .NET, .NET Core, HTML5, CSS3, MsSQL, MySQL, ReactJS, Web services (WSD
L, SOAP, RESTful), Relational databases, Data access/database queries, MVC architectur
Education:  Bachelor's Degree in Computer Science
Experience: 2 years
-----
```

```
Title:      Software Engineer
Category:   Technology/IT
Skills:     REST API, C/C++, Linux/Unix, Python, Go, Git, Gerrit, Jenkins, configuratio
n and deployment of large systems, cloud platforms, VMs, containers, control or data
Education:  Not specified
Experience: Minimum 7 years
```

Save the results

```
In [18]: df.to_csv("../outputs/part2_job_results.csv", index=False)
         print("Saved", len(df), "rows to outputs/part2_job_results.csv")
```

Saved 25 rows to outputs/part2_job_results.csv